

# Project 2 — From Inference to Discovery (Modules 5–7)

## INFO 521 — Machine Learning Foundations

i Course-schedule placement

**Assigned Week 5 • due Week 7.** Milestones on the module arc: 2.1 Approximate inference (M5, Wk 5) → 2.2 Classification/SVM (M6, Wk 6) → 2.3 Clustering (M6, Wk 6) → 2.4 PCA (M7, Wk 7) → 2.5 Synthesis (Wk 7). Gating: Checkpoint 4 (Gate A), Checkpoint 5 (Gate B).

Project 2 continues the *same* clinical cohort from Project 1 and extends it from prediction into discovery. Two things expand at once: the **method** (regression → classification → unsupervised) and the **feature space** (single predictor → the six-feature non-BP set). It is organized as **two thematic gates plus a capstone**.

### Milestones

**Gate A — Supervised** (predict the hypertension label from the non-BP features)

1. **2.1 Approximate Bayesian inference** (M5) — Bayesian logistic regression; from-scratch log-posterior / gradient / Hessian (gradient-checked), Newton–Raphson (MAP), Laplace approximation, Metropolis sampler; MAP/Laplace/Metropolis mapped to P1’s point/likelihood/posterior. Gated by **Checkpoint 4**.
2. **2.2 Classification** (M6) — from-scratch soft-margin SVM (primal hinge + L2, Pegasos), precision/recall/F1, the dual + kernel trick (conceptual), probabilistic-vs-margin showdown on one split.

**Gate B — Unsupervised** (drop the labels; find structure)

3. **2.3 Clustering** (M6) — from-scratch k-means, random restarts for local minima, K selection (elbow / silhouette), and the cluster-vs-held-out-label reveal. Gated by **Checkpoint 5**.
4. **2.4 Dimensionality reduction** (M7) — from-scratch PCA (covariance/SVD), variance explained, biplot, and the k-means clusters re-viewed in PC space.

### Capstone

5. **2.5 Synthesis** — one narrative across the whole semester on one cohort; supervised vs. unsupervised argument; structured peer review.

### The additive bridge

Project 1 ended on a closed-form conjugate Gaussian posterior. Project 2 breaks it deliberately:

1. **Approximate Bayesian inference (M5).** Binarize systolic BP into a **hypertension** label (ACC/AHA:  $SBP \geq 130$  or  $DBP \geq 80$ ) and predict it from the **non-BP** features — the blood-pressure columns are excluded so the label can’t leak into its own predictors. A Bayesian **logistic regression** model has **no conjugate posterior** — so you implement, from scratch: Newton–Raphson (MAP), the Laplace approximation, and a Metropolis sampler. Compare what each buys you.

2. **Classification (M6).** The same binary label is now an SVM target. Soft-margin primal (with the dual and kernel trick derived conceptually). Evaluate with precision / recall / F1. Contrast the discriminative SVM with the probabilistic classifier from step 1.
3. **Clustering (M6).** Drop the labels. Implement k-means from scratch on the six-feature set; address local minima (restarts) and model selection (choosing K). Do the discovered clusters relate to the hypertension label?
4. **Dimensionality reduction (M7).** PCA from scratch on the features; visualize the cohort in 2-D; feed components into the clustering and discuss.

## Gates & bundle (specs)

The Project 2 bundle is **Satisfactory** when:

- **Gate A — Supervised:** 2.1 and 2.2 Satisfactory, and **Checkpoint 4** (Newton–Raphson update) passed.
- **Gate B — Unsupervised:** 2.3 and 2.4 Satisfactory, and **Checkpoint 5** (k-means objective / PCA) passed.
- **Capstone:** 2.5 synthesis Satisfactory (structured peer review completed).

See `../GRADING.md` for the specs mechanics and `../checkpoints/` for the slips.

## Play, extended

- A held-out **prediction round** on the binary outcome (compare SVM vs. Bayesian logistic regression on the same split).
- A **cluster-vs-label reveal**: do unsupervised subgroups recover clinical hypertension strata you never showed the algorithm?
- PCA **biplot exploration**: which features drive the principal axes?

## Carry-over outcomes

The same plumbing applies: `info521.data.hypertension(ds)` provides the leakage-safe Gate A label (ACC/AHA hypertension); `load_clinical()` returns the feature matrix with the blood-pressure columns (`sbp`, `dbp`) excluded, so the Gate A/B feature set is non-BP by construction; `checks.expect_gradient` verifies the 2.1 gradient. No estimators are added to the library — k-means, PCA, SVM, and the samplers are all student-implemented.